

LeetCode Patterns

Complete Study Guide

27 Core Patterns · 160+ Problems · DSA Mastery Roadmap

27	160+	6	Easy→Hard
Patterns	Problems	Difficulty Levels	Progression

Arrays & Strings	Linked Lists	Trees & Graphs	Dynamic Programming	Greedy & Sorting	Advanced DS
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Pattern Reference Cards

1 Sliding Window

Difficulty: Easy→Hard

CONCEPT

Maintain a window over a contiguous subarray/substring. Expand right, shrink left when constraint violated. $O(n)$ time.

SUBTYPES / USE-CASES

- Fixed-size window
- Variable-size window
- Character frequency tracking

LEETCODE PROBLEMS

#3	#76	#209
#424	#567	#904

2 Two Pointers

Difficulty: Easy→Hard

CONCEPT

Use two indices moving toward each other or in the same direction to reduce $O(n^2)$ brute force to $O(n)$.

SUBTYPES / USE-CASES

- Opposite ends (sorted array)
- Same direction (partitioning)
- Container / area problems

LEETCODE PROBLEMS

#11	#15	#16
#18	#42	#167

3 Fast/Slow Pointers

Difficulty: Easy→Medium

CONCEPT

Two pointers at different speeds to detect cycles or find midpoints in linked lists. Floyd's cycle detection.

SUBTYPES / USE-CASES

- Cycle detection
- Cycle entry point
- Middle of list
- Intersection

LEETCODE PROBLEMS

#141	#142	#19
#876	#160	#234

4

Binary Search on Sorted Data

Difficulty: Easy→Medium

CONCEPT

Eliminate half the search space each iteration on sorted or rotated sorted arrays. $O(\log n)$ time.

SUBTYPES / USE-CASES

- Classic search
- Leftmost/rightmost position
- Rotated sorted array
- Peak finding

LEETCODE PROBLEMS

#33	#34	#35
#153	#162	#704

5

Binary Search on Answer

Difficulty: Medium→Hard

CONCEPT

When the answer lives in a monotonic range, binary search on the answer space and use a feasibility check.

SUBTYPES / USE-CASES

- Minimize maximum
- Maximize minimum
- Capacity / allocation problems

LEETCODE PROBLEMS

#875	#1011	#410
#774	#1283	#1482

6

Hashing / Frequency Maps

Difficulty: Easy→Medium

CONCEPT

Use hash maps/sets for $O(1)$ lookup, grouping, counting, or de-duplication. Trade space for time.

SUBTYPES / USE-CASES

- Two-sum lookup
- Anagram grouping
- Longest consecutive sequence
- Top-K frequency

LEETCODE PROBLEMS

#1	#49	#128
#217	#242	#347

7

Prefix Sum / Running Sum

Difficulty: Easy→Medium

CONCEPT

Precompute cumulative sums so any subarray sum is $O(1)$. Combine with hash maps for subarray sum targets.

SUBTYPES / USE-CASES

- Range sum query
- Subarray sum equals K
- Pivot index
- Divisible subarray

LEETCODE PROBLEMS

#303	#560	#724
#930	#974	#523

8

Difference Array / Range Updates

Difficulty: Medium

CONCEPT

Apply range increment/decrement in $O(1)$ using a difference array, then reconstruct with prefix sum.

SUBTYPES / USE-CASES

- Range add queries
- Interval coverage
- Flower planting
- Car pooling

LEETCODE PROBLEMS

#370	#1094	#1109
#1893	#1943	#2381

9

Monotonic Stack

Difficulty: Medium→Hard

CONCEPT

Maintain a stack in strictly increasing or decreasing order to answer next-greater/smaller in $O(n)$ amortized.

SUBTYPES / USE-CASES

- Next greater element
- Largest rectangle in histogram
- Stock span
- Maximal rectangle

LEETCODE PROBLEMS

#739	#496	#503
#84	#85	#901

10

Monotonic Queue / Deque

Difficulty: Hard

CONCEPT

Deque-based window max/min in $O(n)$. Front holds the optimal element; expired elements are popped from front.

SUBTYPES / USE-CASES

- Sliding window maximum
- Constrained subsequence sum
- Jump game with deque

LEETCODE PROBLEMS

#239	#862	#1425
#1438	#1499	#1696

11

Heap / Top K

Difficulty: Medium

CONCEPT

Use a min-heap of size K for Top-K elements in $O(n \log K)$. Max-heap for kth-smallest variants.

SUBTYPES / USE-CASES

- Kth largest element
- Top K frequent
- K closest points
- Last stone weight

LEETCODE PROBLEMS

#215	#347	#692
#703	#973	#1046

12

Intervals

Difficulty: Easy→Hard

CONCEPT

Sort by start time. Merge overlapping intervals, count rooms needed, or find minimum arrows.

SUBTYPES / USE-CASES

- Merge intervals
- Insert interval
- Meeting rooms
- Non-overlapping intervals

LEETCODE PROBLEMS

#56	#57	#252
#253	#435	#452

13

Greedy Scheduling / Sorting

Difficulty: Medium→Hard

CONCEPT

Make locally optimal choices. Often requires a clever sort key or greedy observation about the optimal structure.

SUBTYPES / USE-CASES

- Jump game
- Task scheduler
- Queue reconstruction
- Partition labels

LEETCODE PROBLEMS

#45	#55	#406
#621	#763	#134

14

Linked List Manipulation

Difficulty: Easy→Hard

CONCEPT

In-place pointer rewiring. Use dummy head nodes, prev/curr pointers, and careful order of assignments.

SUBTYPES / USE-CASES

- Merge sorted lists
- Reverse sublist
- Swap in pairs
- Deep copy with random

LEETCODE PROBLEMS

#21	#23	#24
#25	#92	#138

15

Tree DFS

Difficulty: Easy→Hard

CONCEPT

Recursive or iterative pre/in/post-order traversal. Pass state top-down; aggregate results bottom-up.

SUBTYPES / USE-CASES

- Max depth
- Path sum
- Diameter
- Max path sum
- Mirror tree

LEETCODE PROBLEMS

#104	#112	#113
#543	#124	#226

16

Tree BFS / Level Order

Difficulty: Easy→Medium

CONCEPT

Use a queue to process nodes level by level.
Track level boundaries with size snapshotting.

SUBTYPES / USE-CASES

- Level order traversal
- Zigzag
- Right side view
- Average of levels

LEETCODE PROBLEMS

#102	#103	#199
#515	#637	#116

17

BST Problems

Difficulty: Easy→Hard

CONCEPT

Leverage BST invariant (left < root < right).
In-order traversal yields sorted order.

SUBTYPES / USE-CASES

- Validate BST
- Recover BST
- Kth smallest
- LCA
- Delete node
- Search

LEETCODE PROBLEMS

#98	#99	#230
#235	#450	#700

18

Backtracking Basics

Difficulty: Medium

CONCEPT

Explore all combinations/permutations via DFS +
undo. Prune early to cut branches.

SUBTYPES / USE-CASES

- Permutations
- Combinations
- Subsets
- Combination sum

LEETCODE PROBLEMS

#46	#47	#77
#78	#90	#39

19

Backtracking with Constraints

Difficulty: Medium→Hard

CONCEPT

Backtracking with tighter pruning: used-element tracking, board constraints, or partition rules.

SUBTYPES / USE-CASES

- Combination sum II
- Phone letters
- Word search
- N-Queens
- Palindrome partitioning

LEETCODE PROBLEMS

#40	#17	#79
#131	#51	#52

20

Graph BFS / DFS

Difficulty: Easy→Medium

CONCEPT

Traverse grids or adjacency lists. BFS for shortest path; DFS for connected components.

SUBTYPES / USE-CASES

- Number of islands
- Flood fill
- Rotting oranges
- Shortest path in binary matrix

LEETCODE PROBLEMS

#200	#695	#733
#994	#1091	#1254

21

Topological Sort / DAG

Difficulty: Medium→Hard

CONCEPT

Kahn's algorithm (BFS in-degree) or DFS post-order to linearize a DAG. Detect cycles simultaneously.

SUBTYPES / USE-CASES

- Course schedule
- Find eventual safe states
- Sort items by groups

LEETCODE PROBLEMS

#207	#210	#802
#1462	#1203	#2115

22

Union Find / DSU

Difficulty: Medium→Hard

CONCEPT

Disjoint Set Union with path compression + union by rank. Near- $O(1)$ amortized find/union.

SUBTYPES / USE-CASES

- Number of provinces
- Redundant connection
- Smallest string with swaps

LEETCODE PROBLEMS

#547	#684	#1319
#1579	#990	#1202

23

Shortest Path

Difficulty: Medium→Hard

CONCEPT

Dijkstra (non-negative weights), Bellman-Ford (negative edges), BFS (unit weights), or modified variants.

SUBTYPES / USE-CASES

- Dijkstra
- Cheapest flights (K stops)
- Path with max probability
- Swim in rising water

LEETCODE PROBLEMS

#743	#787	#1514
#1631	#1334	#1976

24

MST / Graph Greedy

Difficulty: Hard

CONCEPT

Kruskal's (sort edges + DSU) or Prim's (priority queue) for minimum spanning tree.

SUBTYPES / USE-CASES

- Min cost to connect points
- Optimize water distribution
- Swim in rising water

LEETCODE PROBLEMS

#1584	#1135	#1168
#1489	#778	#1102

25 Trie

Difficulty: Medium→Hard

CONCEPT

Prefix tree for $O(L)$ insert/search where L = word length. Enables prefix queries and autocomplete.

SUBTYPES / USE-CASES

- Implement trie
- Word search II
- Replace words
- Map sum pairs

LEETCODE PROBLEMS

#208	#211	#212
#648	#677	#1268

26 Bit Manipulation

Difficulty: Easy→Medium

CONCEPT

XOR, AND, OR, shifts to solve problems in $O(1)$ space. XOR cancels duplicates; bit counting via kernighan.

SUBTYPES / USE-CASES

- Single number
- Missing number
- Counting bits
- Reverse bits

LEETCODE PROBLEMS

#136	#137	#191
#338	#268	#190

27 1D DP Basics

Difficulty: Easy→Medium

CONCEPT

Build solutions from smaller subproblems. Define state, recurrence, and base cases clearly.

SUBTYPES / USE-CASES

- Climbing stairs
- House robber
- Coin change
- Perfect squares
- LIS

LEETCODE PROBLEMS

#70	#198	#213
#322	#279	#300